Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

Department of Software Engineering

OPERATING SYSTEMS AND SYSTEM PROGRAMMING: HARDWARE ECOSYSTEM ANALYSIS

Section-B

Group-4

Group Members	ID Number
1.BIRUH BIZUALEM	BDU1701727
2.HABTAMU FIKADU	BDU1702071
3.ELDANA GULMA	BDU1701854
4.GIDEON BERHANU	BDU1604303
5.EYERUSALEM GETAHUN	BDU1701933
6. BINIYAM DESALEGN	BDU1701709

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1. Introduction

Modern computing systems are built on a layered architecture where hardware components and software systems work in continuous coordination to achieve efficient, reliable, and high-performance computing. At the center of this ecosystem is the Operating System (OS), which functions as a critical intermediary between physical hardware and user-level applications.

The OS is responsible for managing and coordinating core system resources, including:

- Processors (CPU) for instruction execution
- Memory (RAM) for temporary data storage and fast access
- Storage devices (SSD/HDD) for permanent data retention
- Input/Output devices for user interaction

In essence, the Operating System ensures that hardware resources are allocated efficiently, processes are executed in an organized manner, and system stability is maintained even under heavy computational load.

From a system programming perspective, the OS provides a controlled environment through system calls, allowing applications to interact with hardware without directly accessing low-level components. This abstraction improves security, portability, and stability of computing systems.

1.1 Role of Hardware–Software Integration

Modern systems depend heavily on tight integration between hardware and software layers:

- Hardware provides the physical computation and storage capabilities
- The OS provides resource management and coordination
- System programs provide low-level control and optimization tools

Without this integration:

- Applications would not be able to access hardware safely
- Memory usage would become unstructured
- CPU tasks would not be efficiently scheduled
- System performance would degrade significantly

1.2 Hardware Companies Under Study

This report analyzes four major technology companies, each representing a key layer of computing architecture:

1. MediaTek (Processor Systems)

MediaTek is responsible for designing System-on-Chip (SoC) processors that combine CPU, GPU, and AI processing units into a single integrated circuit. These chips are widely used in mobile and embedded systems, where efficiency and performance must be balanced.

2. ASUS (Computer Systems)

ASUS manufactures motherboards, laptops, and desktop systems that physically connect all hardware components. It plays a key role in system initialization through BIOS/UEFI firmware and supports hardware expansion and compatibility.

3. Corsair (Memory Systems)

Corsair specializes in DRAM and DIMM memory modules, which provide high-speed temporary storage for active processes. RAM is essential for multitasking and directly impacts system performance and responsiveness**.**

4. Kioxia (Storage Systems)

Kioxia develops NAND flash memory and SSD technologies, which are responsible for permanent data storage. These devices ensure data persistence and fast retrieval in modern computing systems.

1.3 Objective of the Report

The primary objective of this study is to:

- Analyze how each hardware component contributes to overall system functionality
- Understand how the Operating System manages and coordinates these hardware resources
- Examine the role of system programming concepts such as system calls, memory management, and process scheduling
- Evaluate the advantages and limitations of each hardware technology in real-world computing environments

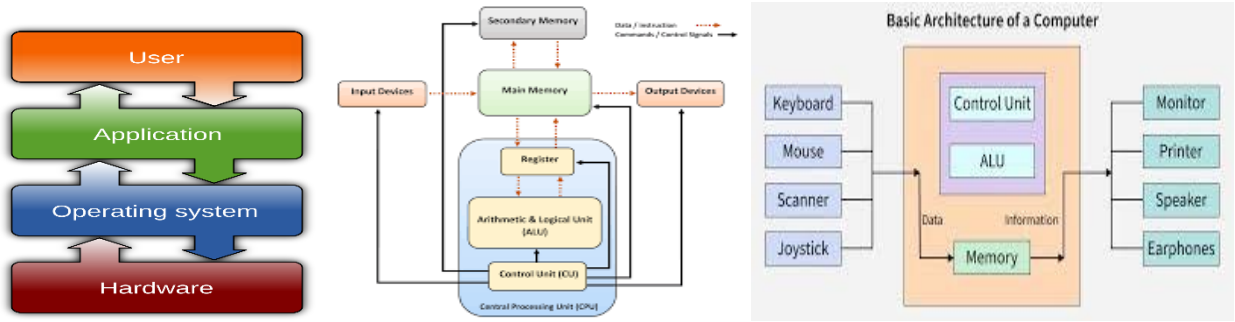
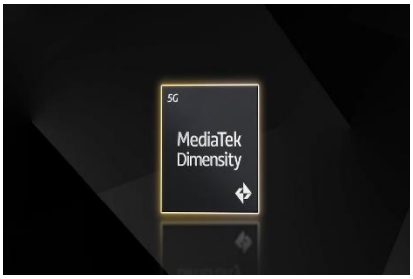
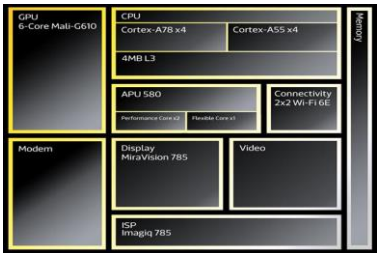


Figure 1.3 Computer System Block Diagram

2. Processor Systems (MediaTek)

Overview





MediaTek

MediaTek is a Taiwanese semiconductor company that designs system-on-chip (SoC) solutions for mobile devices, home entertainment, connectivity, and Internet of Things (IoT) products. It is one of the world's largest fabless chip designers, competing with major players such as Qualcomm and Samsung Electronics in the global market.

Key facts

- **Founded:** 1997 (spun off from United Microelectronics Corporation)
- **Headquarters:** Hsinchu, Taiwan
- **CEO:** Rick Tsai
- **Main brands:** Dimensity (5G mobile chips), Helio (4G mobile chips)
- **Ticker:** 2454 (Taiwan Stock Exchange)

History and growth

MediaTek originated as a spin-off from United Microelectronics Corporation's consumer electronics division, focusing initially on optical drives and digital TVs. In the mid-2000s, it expanded into mobile chipsets, becoming a key supplier for emerging-market handset manufacturers. The company's affordable, power-efficient designs enabled rapid adoption among Chinese and Indian smartphone brands.

Products and technologies

MediaTek's portfolio spans mobile processors, home entertainment SoCs, and connectivity chips. The Dimensity series targets flagship and mid-range 5G smartphones, integrating AI, imaging, and modem functions. Beyond phones, MediaTek chips power smart TVs, routers, Chromebooks, and automotive electronics. The company also develops AI and Wi-Fi 7 solutions for next-generation connectivity.

Market position and partnerships

MediaTek consistently ranks among the top global smartphone SoC vendors by shipment volume, often alternating with Qualcomm for the top spot. Its chips are used by major OEMs including Xiaomi, Oppo, Vivo, and Samsung. The firm's strategy emphasizes high integration, cost efficiency, and close alignment with regional supply chains.

Research and innovation

The company invests heavily in 5G modem development, AI processing, and energy-efficient design. MediaTek maintains multiple R&D centers worldwide, supporting advancements in mobile computing, wireless communication, and multimedia processing. Its innovation focus aims to extend beyond smartphones into automotive, smart home, and industrial IoT ecosystems.

Architecture Explanation

MediaTek utilizes the **ARMv9 Architecture**, employing a **Heterogeneous Multi-Processing (HMP)** design known as **Cortex-X Custom cores**. Their architecture follows a 1+3+4 structure: one ultra-large core for peak performance, three large cores for sustained tasks, and four efficiency cores for background processes.

OS Interaction

The OS kernel interacts with MediaTek processors through **Energy-Aware Scheduling (EAS)**. The kernel must be aware of the "cost" of waking up a high-performance core versus an efficiency core. The OS uses **Dynamic Voltage and Frequency Scaling (DVFS)** to adjust power based on the workload.

System Programming Perspective

System programmers work with the **Instruction Set Architecture (ISA)**—specifically ARM64. Low-level programming involves optimizing code for the **NEON SIMD (Single Instruction, Multiple Data)** engine to accelerate multimedia tasks. Programmers must also manage **Interrupt Request (IRQ)** affinity to ensure high-priority hardware interrupts are handled by the most responsive core.

- **Advantages:** Exceptional power-to-performance ratio; integrated 5G and AI hardware.
- **Limitations:** Thermal throttling under sustained heavy loads; closed-source driver blobs can hinder custom kernel development.
- **Key Insight:** MediaTek's success lies in its ability to offload OS tasks from the CPU to specialized sub-processors (like the APU), reducing general-purpose CPU cycles.

3. Computer Systems (ASUS)

Overview

ASUS

ASUS (officially **ASUSTeK Computer Inc.**) is a Taiwanese multinational technology company specializing in computer hardware, electronics, and IT solutions. Known for its innovation in PCs, laptops, and motherboards, ASUS has become one of the world’s largest computer manufacturers and a key brand in gaming and consumer electronics.

Key facts

- **Founded:** 1989, Taipei, Taiwan
- **Founders:** T.H. Tung, Ted Hsu, Wayne Hsieh, M.T. Liao

- **Headquarters:** Beitou District, Taipei, Taiwan
- **CEO (2024):** S.Y. Hsu and Samson Hu (co-CEOs)
- **Brands:** Republic of Gamers (ROG), TUF Gaming, ZenBook, VivoBook

Origins and Growth

ASUS was established by four former engineers from Acer Inc. who aimed to build high-quality motherboards. Its early success in motherboard design helped the company become a leading supplier for major chipsets. Through the 1990s and 2000s, ASUS expanded into full computer systems, monitors, networking devices, and mobile phones.

Product Portfolio

ASUS's portfolio spans personal and gaming laptops, desktops, smartphones, PC components, and peripherals. The **ROG (Republic of Gamers)** line targets high-performance gaming, while **ZenBook** and **VivoBook** focus on premium and consumer markets, respectively. ASUS is also a major player in networking hardware and computer components like graphics cards and motherboards.

Innovation and Recognition

ASUS has received numerous design and innovation awards for its products, particularly in the gaming and ultraportable laptop segments. It was among the first companies to popularize netbooks with the Eee PC and continues to integrate cutting-edge features such as OLED displays, dual screens, and AI-driven performance optimization.

Market Position and Impact

As of the 2020s, ASUS ranks among the top five PC vendors globally by shipment volume. Its ROG brand holds a strong presence in esports and gaming hardware markets. Beyond consumer products, ASUS invests in cloud computing, AI, and sustainable manufacturing practices to diversify and modernize its technology offerings.

Architecture Explanation

ASUS architecture focuses on the **Motherboard and Chipset (PCH)**. They design the PCB traces that connect the CPU to the RAM and PCIe lanes. Crucially, ASUS develops the **UEFI (Unified Extensible Firmware Interface)**, which is the first code the OS loader encounters.

OS Interaction

The interaction occurs primarily through the **ACPI (Advanced Configuration and Power Interface)** tables. ASUS firmware tells the OS how many cores exist, what the thermal limits are, and how to manage sleep states (S3/S4). The OS uses **drivers** to communicate with ASUS-specific hardware like the Aura Lighting controller or the Armoury Crate interface.

System Programming Perspective

System programming here involves **Firmware Development** and **Kernel Module** writing. Programmers interact with the **SMBus** and **I2C** protocols to read sensor data (fan speed, voltage). Developing "Fan Control" software requires writing to specific I/O ports defined in the ASUS hardware specification.

- **Advantages:** High-quality power delivery (VRMs); extensive BIOS customization for overclocking.
- **Limitations:** Proprietary software "bloatware" can sometimes conflict with standard Windows/Linux kernel scheduling.
- **Key Insight:** ASUS acts as the "Architect," where the OS relies on the UEFI to understand the physical limits of the machine.

4. Memory Systems (Corsair)

Overview

Corsair

Corsair (formally Corsair Gaming, Inc.) is an American computer hardware and peripherals company focused on high-performance products for PC gamers, content creators, and enthusiasts. It designs and sells components like memory and power supplies alongside gaming gear such as keyboards, mice, headsets, and streaming equipment.

Key facts

- **Founded:** 1994, Fremont, California, USA
- **Headquarters:** Milpitas, California, USA

- **Public listing:** NASDAQ: CRSR; S&P 600 component
- **Employees:** About 2,500+ (mid-2020s)
- **Notable brands/subsidiaries:** Elgato, Origin PC, SCUF Gaming, Drop, Fanatec

Background and history

Corsair started in 1994 as Corsair Microsystems, initially producing cache modules for OEMs. When CPU designs integrated cache, the company pivoted to DRAM modules, first for servers and later for enthusiasts and overclockers. High-speed, overclockable RAM lines like Vengeance and Dominator Platinum became its signature products.

Over time Corsair expanded to power supplies, cases, cooling (air and liquid), and solid-state storage, then into gaming peripherals and full systems. Its IPO in 2020 formalized its position as a major gaming-hardware brand.

Product ecosystem

Corsair now sells a broad ecosystem designed to interoperate via its iCUE software, including:

- PC components: DRAM, PSUs, cases, fans, coolers, SSDs
- Gaming peripherals: mechanical keyboards, mice, headsets, mousepads, speakers
- Creator gear: capture cards, stream decks, lights, and mics through Elgato
- Systems and specialty hardware: prebuilt gaming PCs (Origin PC), custom controllers (SCUF), enthusiast keyboards and audio (Drop), and sim-racing gear (Fanatec product line).

Business model and market position

Corsair operates a mixed hardware–software–services model, selling through retailers, e-tailers, its own online store, and B2B channels. Revenue is split broadly between “Gamer and Creator Peripherals” and “Gaming Components and Systems,” with components like memory and PSUs historically forming a large share.

The company is considered one of the leading global gaming-hardware manufacturers, with strong market share in enthusiast DRAM and mechanical keyboards, and a sizable presence in streaming gear via Elgato and related brands.

Recent developments

In the mid-2020s Corsair has continued broadening its portfolio via acquisitions (notably Drop and Fanatec’s product line) and has navigated challenging component markets such as DRAM and semiconductors. It has also updated packaging and security features on high-end DDR5 RAM to reduce retail fraud, reflecting its significant stake in the memory market.

Architecture Explanation

Corsair modules (like Vengeance or Dominator) utilize high-bin DRAM chips. The architecture includes the **SPD (Serial Presence Detect)** chip, which stores timing data ($CL, tRCD, tRP, tRAS$) and **XMP (Extreme Memory Profile)** data for automated overclocking.

OS Interaction

The OS manages Corsair RAM through **Paging** and **Virtual Memory Management**. The kernel maintains a **Page Table** that maps a program's virtual addresses to the physical addresses on the Corsair DIMM. The OS also handles **ECC (Error Correction Code)** if the specific Corsair module supports it, logging hardware memory errors to the system log.

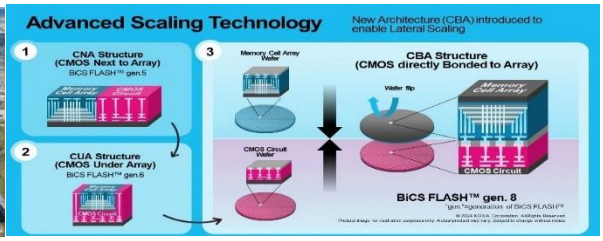
System Programming Perspective

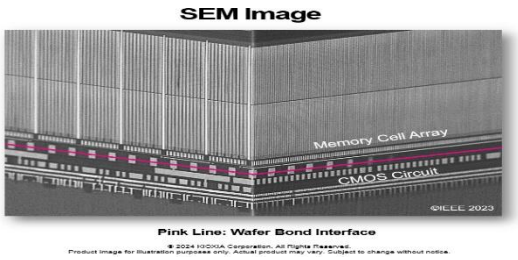
Programmers optimize for Corsair RAM by focusing on **Cache Locality**. Since RAM access is orders of magnitude slower than CPU cache, system programmers use **Data-Oriented Design** to ensure data is fetched in contiguous blocks, minimizing "Page Faults" and maximizing the high bandwidth (MT/s) Corsair provides.

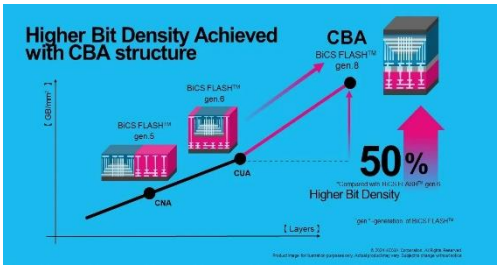
- **Advantages:** High frequency and low latency; integrated heat spreaders for thermal stability.
- **Limitations:** High-speed XMP profiles can sometimes cause system instability if the CPU's memory controller is weak.
- **Key Insight:** High-performance RAM is useless without an OS kernel that efficiently manages the Translation Lookaside Buffer (TLB).

5. Storage Systems (Kioxia)

Overview







Kioxia

Kioxia Corporation is a Japanese multinational company specializing in flash memory and solid-state drive (SSD) technology. Headquartered in Tokyo, it is among the world's largest producers of NAND flash, continuing the legacy of innovation that began under its predecessor, Toshiba Memory Corporation**.**

Key facts

- **Founded: April 1, 2017 (as Toshiba Memory Corporation)**
- **Rebranded: October 1, 2019**
- **Headquarters: Minato-ku, Tokyo, Japan**
- **President & CEO: Nobuo Hayasaka**
- **Employees: ~15,000 (consolidated, 2025)**
- **Parent company: Kioxia Holdings Corporation**

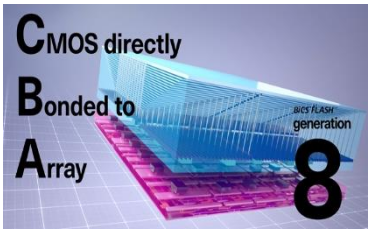
Origins and rebranding

Kioxia was created in 2017 when Toshiba Corporation spun off its memory business. In 2019, the company adopted the name "Kioxia," blending the Japanese word *kioku* ("memory") with the Greek *axia* ("value"). The rebranding reflected its mission to "uplift the world with memory" while maintaining its technological leadership in flash storage.

Technological leadership

Kioxia is credited with inventing NAND flash memory in 1987 and remains a core innovator in storage. Its signature BiCS FLASH™ 3D NAND technology underpins applications in smartphones, data centers, and AI infrastructure. The company also collaborates with Western Digital on joint NAND fabrication and develops emerging memory types such as MRAM and storage-class memory.

Manufacturing and facilities



Kioxia operates some of the world's largest flash fabrication facilities, notably its Yokkaichi Plant in Mie Prefecture and the newer Kitakami Plant in Iwate Prefecture. These advanced sites employ AI-driven processes to manufacture 3D NAND with high yield and energy efficiency.

Global presence and market role

The company maintains subsidiaries across Asia, Europe, and the Americas, supplying enterprise and consumer SSDs, memory cards, and embedded storage. Holding roughly 18% of the global NAND

market, Kioxia ranks among top competitors like Samsung Electronics, SK Hynix, and Micron Technology. Despite market volatility, it remains central to the semiconductor ecosystem driving 5G, AI, and cloud computing. (KIOXIA Corporation)

Architecture Explanation

Kioxia uses BiCS FLASH (3D NAND) technology, stacking memory cells vertically to increase density. Their SSDs utilize the NVMe (Non-Volatile Memory Express) protocol over PCIe lanes, managed by a sophisticated on-board controller.

OS Interaction

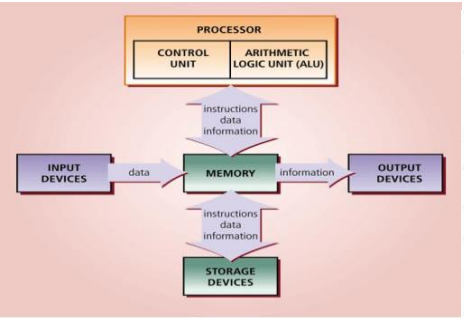
The OS communicates with Kioxia drives via the **Block I/O Subsystem**. When a program calls read(), the OS sends an asynchronous request to the NVMe queue. The OS also manages **TRIM commands**, telling the Kioxia controller which blocks of data are no longer in use to maintain write speeds.

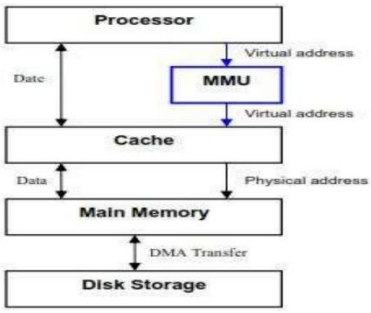
System Programming Perspective

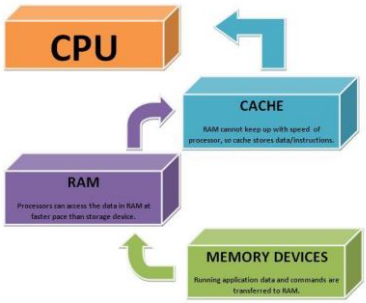
Writing for Kioxia drives involves understanding **Direct Memory Access (DMA)**. System programmers write drivers that allow the SSD to transfer data directly to Corsair RAM without involving the CPU, significantly reducing latency. They also implement **Wear Leveling** logic in firmware to prevent specific NAND cells from burning out.

- **Advantages:** Massive IOPS (Input/Output Operations Per Second); lower power consumption than HDDs.
- **Limitations:** Finite write endurance (TBW - Total Bytes Written); performance degrades as the drive reaches capacity.
- **Key Insight:** The OS view of storage has shifted from "spinning platters" to "memory-mapped files" due to Kioxia's high-speed flash.

6. System Integration (Critical Analysis)







Modern computing performance depends on how effectively hardware components work together under the control of the Operating System. Rather than functioning independently, CPU, memory,

storage, and motherboard form a tightly connected ecosystem where performance is determined by overall system balance.

6.1 Data Flow Structure

In a typical execution cycle, data follows this path:

Storage (Kioxia) → RAM (Corsair) → CPU (MediaTek) → Output

Kioxia provides persistent storage through SSD technology. When a program is executed, the OS loads it into Corsair RAM for fast access, and MediaTek processors execute instructions.

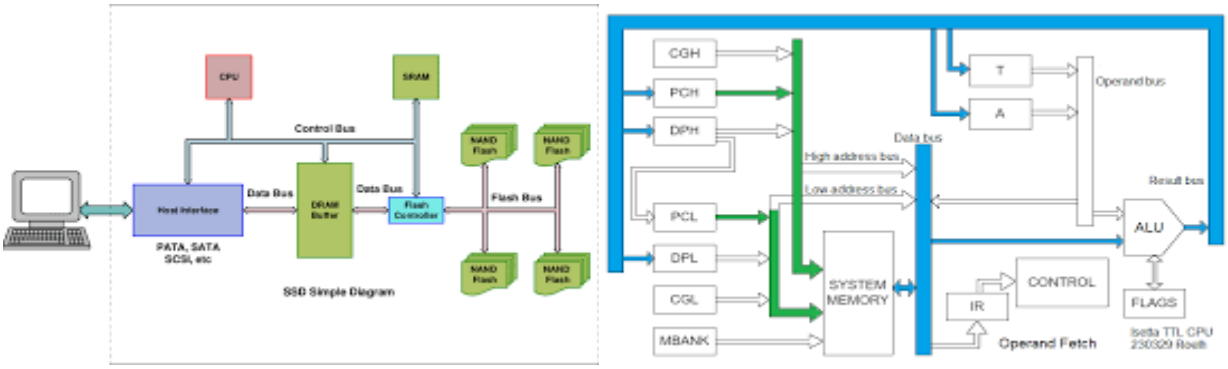


Figure 6.1 **Storage → RAM → CPU Data Flow Diagrams**

6.2 System Bottlenecks

Even high-performance systems face limitations:

- **Motherboard limitation:** ASUS bus speed can restrict data transfer between components
- **Memory latency:** CPU may wait for RAM data despite high processing speed
- **Storage I/O delay:** SSD speed depends on OS scheduling and controller efficiency

A slowdown in any layer affects the entire system.

6.3 Thermal and Power Control

High-performance components generate heat, especially CPU and SSD under heavy workloads. The system firmware (BIOS/UEFI) and OS manage:

- Temperature monitoring
- Voltage regulation
- Fan speed control
- Performance throttling when necessary

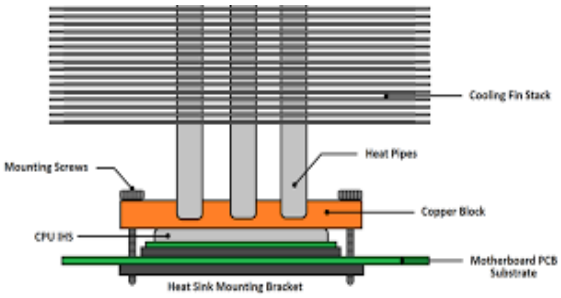


Figure 6.3 **Thermal Management System Diagrams**

6.4 Kernel-Level Role

The Operating System kernel ensures efficient coordination:

- Manages CPU scheduling for MediaTek processors
- Handles fast I/O requests from Kioxia SSDs
- Controls memory allocation in Corsair RAM

Modern kernels (Windows/Linux) are optimized for parallel processing, virtual memory, and high-speed storage access.

6.5 Key Insight

System performance is determined not by a single component, but by the **efficiency of interaction between all components**. The OS plays a central role in balancing speed, power, and resource allocation to prevent bottlenecks.

6.6 Summary Limitations

- Hardware bottlenecks (bus speed, memory latency)
- Thermal constraints under heavy processing
- OS overhead in resource management
- Dependency on driver and firmware efficiency

Final Insight

A modern computing system achieves optimal performance only when CPU, memory, storage, and motherboard operate in synchronized harmony under intelligent OS control.

7. Comparative Analysis

Feature	MediaTek (CPU)	ASUS (System)	Corsair (RAM)	Kioxia (SSD)
Primary Role	Computation	Integration	Volatile Workspace	Persistence
OS Interface	ISA / Scheduler	UEFI / ACPI	Page Tables	Block I/O / NVMe
Programming	Assembly / C	C++ / Firmware	Memory Mapping	DMA / Drivers
Critical Metric	IPC / Watts	Thermal Headroom	Latency (ns)	IOPS

8. Future Trends

The evolution of computing systems is moving toward deeper integration between hardware intelligence and Operating System optimization. Future developments aim to reduce bottlenecks, improve parallel processing, and enhance predictive computing capabilities.

8.1 Compute-in-Memory Architecture

In next-generation systems, memory modules such as those developed by Corsair may include embedded processing capabilities.

Instead of only storing data, these advanced memory units will:

- Perform simple computations inside the memory chip
- Reduce data movement between RAM and CPU
- Improve overall system efficiency and speed

This reduces dependency on the CPU (e.g., MediaTek processors) for basic data operations, significantly lowering latency.

8.2 DirectStorage Optimization

Future Operating Systems will increasingly adopt technologies like DirectStorage, where storage devices communicate more directly with the GPU.

High-speed SSDs from Kioxia will:

- Send compressed game or application data directly to the GPU
- Bypass the CPU for certain data transfer tasks
- Eliminate traditional I/O bottlenecks

This improves loading times and supports high-performance applications such as gaming and real-time rendering.

8.3 AI-Driven Operating Systems

Next-generation systems will integrate artificial intelligence at the OS level, often powered by AI-capable processors from MediaTek.

These intelligent systems will:

- Predict user behavior based on usage patterns
- Pre-load frequently used applications into Corsair RAM
- Optimize data retrieval from Kioxia storage

This leads to a more responsive and adaptive computing experience.

9.Conclusion

This documentation underscores that modern computing hardware operates within a dynamic and continuously advancing ecosystem rather than as isolated components. Each element fulfills a distinct and strategic function within the overall architecture:

- MediaTek serves as the computational core, responsible for executing instructions and handling processing tasks
- ASUS delivers the foundational infrastructure that integrates and interconnects system components
- Corsair provides high-performance volatile memory to support real-time processing demands
- Kioxia enables reliable long-term data storage and efficient retrieval

At the center of this ecosystem, the Operating System—reinforced by system programming methodologies—functions as the primary orchestrator. It manages inter-component communication, optimizes resource allocation, and ensures seamless execution through effective scheduling, memory management, and data flow control.

In essence, system programming represents the enabling layer that aligns these diverse hardware components into a cohesive and high-performing system, ultimately delivering a computing environment that is responsive, stable, and efficient for end users.

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